Position-dependent motion gain with an unknown wall position

1 Model and estimator

Motion gain as a function of the distance from the wall

$$
\begin{aligned}
h[k] &= \bar{x}_d[k] - \delta\bar{x}_3[k] - \bar{w}_s \\
\bar{a}'[k] &= \frac{1-c}{h[k]^2} \\
M[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\bar{x}_3[k]
\end{aligned}
$$

State of the process

$$
\begin{aligned}
\delta\bar{x}_1[k+1] &= \delta\bar{x}_2[k] \\
\delta\bar{x}_2[k+1] &= \delta\bar{x}_3[k] \\
\delta\bar{x}_3[k+1] &= \lambda_c\,\delta\bar{x}_3[k] - F_{dx}[k]\,e_{\bar{a}}[k] - \varepsilon_x[k] \\
\bar{a}[k+1] &= \bar{a}[k] + \bar{a}'[k]\left(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\right) \\
c[k+1] &= c[k] \\
\bar{w}_s[k+1] &= \bar{w}_s[k]
\end{aligned}
$$

Measurement and output matrix

$$
\mathbf{y}[k] = \begin{bmatrix} \delta\bar{x}_m[k] \\ \bar{a}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{x}}_1[k] \\ \hat{\bar{a}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}
$$

Estimator

$$
\begin{aligned}
\hat{h}[k] &= \bar{x}_d[k] - \delta\hat{\bar{x}}_3[k] - \hat{\bar{w}}_s[k] \\
\hat{\bar{a}}'[k] &= \frac{1-\hat{c}[k]}{\hat{h}[k]^2} \\
\hat{M}[k] &= \Delta\bar{x}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{x}}_3[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{x}}_1[k+1] &= \delta\hat{\bar{x}}_2[k] + \left\{\ell_{11}[k]\,e_{x_1}[k] + \ell_{12}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_2[k+1] &= \delta\hat{\bar{x}}_3[k] + \left\{\ell_{21}[k]\,e_{x_1}[k] + \ell_{22}[k]\,e_{a_m}[k]\right\} \\
\delta\hat{\bar{x}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{x}}_3[k] + \left\{\ell_{31}[k]\,e_{x_1}[k] + \ell_{32}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{a}}[k+1] &= \hat{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\hat{M}[k] + \left\{\ell_{41}[k]\,e_{x_1}[k] + \ell_{42}[k]\,e_{a_m}[k]\right\} \\
\hat{c}[k+1] &= \hat{c}[k] + \left\{\ell_{51}[k]\,e_{x_1}[k] + \ell_{52}[k]\,e_{a_m}[k]\right\} \\
\hat{\bar{w}}_s[k+1] &= \hat{\bar{w}}_s[k] + \left\{\ell_{61}[k]\,e_{x_1}[k] + \ell_{62}[k]\,e_{a_m}[k]\right\}
\end{aligned}
$$

Dynamics of estimation errors

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta\bar{x}_1}[k] \\ e_{\delta\bar{x}_2}[k] \\ e_{\delta\bar{x}_3}[k] \\ e_{\bar{a}}[k] \\ e_{c}[k] \\ e_{\bar{w}_s}[k] \end{bmatrix}, \qquad \mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & 0 & 0 \\ 0 & 0 & \hat{\bar{a}}'[k]\,(1-\lambda_c) + \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] & 1 + \hat{\bar{a}}'[k]\,F_{dx}[k] & -\dfrac{\hat{M}[k]}{\hat{h}[k]^2} & \dfrac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k] \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_x[k] \\ \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ 0 \\ 0 \end{bmatrix}$$

Row 4 of $\mathbf{F}_e$, expanded

(a) the two gain steps and their difference

$$\bar{a}[k+1] - \bar{a}[k] = \bar{a}'[k]\,\big(M[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\big)$$

$$\hat{\bar{a}}[k+1] - \hat{\bar{a}}[k] = \hat{\bar{a}}'[k]\,\hat{M}[k]$$

$$e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = \bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] + \bar{a}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \bar{a}'[k]\,\varepsilon_x[k]$$

(b) each factor written as its estimate plus its error

$$h[k] = \hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k], \qquad c = \hat{c}[k] + e_c[k]$$

$$\bar{a}'[k] = \frac{1 - \hat{c}[k] - e_c[k]}{\big(\hat{h}[k] - e_{\delta\bar{x}_3}[k] - e_{\bar{w}_s}[k]\big)^2} = \hat{\bar{a}}'[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big) - \frac{1}{\hat{h}[k]^2}\,e_c[k] + O(e^2)$$

$$M[k] = \hat{M}[k] + (1 - \lambda_c)\,e_{\delta\bar{x}_3}[k]$$

(c) the product difference, exact

$$\bar{a}'[k]\,M[k] - \hat{\bar{a}}'[k]\,\hat{M}[k] = \hat{\bar{a}}'[k]\,\big(M[k] - \hat{M}[k]\big) + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\,\hat{M}[k] + \big(\bar{a}'[k] - \hat{\bar{a}}'[k]\big)\big(M[k] - \hat{M}[k]\big)$$

(d) every term after substituting (b)

$$\begin{aligned} e_{\bar{a}}[k+1] - e_{\bar{a}}[k] = {} & \hat{\bar{a}}'[k]\,(1 - \lambda_c)\,e_{\delta\bar{x}_3}[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\delta\bar{x}_3}[k] - \frac{\hat{M}[k]}{\hat{h}[k]^2}\,e_c[k] + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,F_{dx}[k]\,e_{\bar{a}}[k] + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \\ & + \Big(\frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\big(e_{\delta\bar{x}_3}[k] + e_{\bar{w}_s}[k]\big) - \frac{e_c[k]}{\hat{h}[k]^2}\Big) \\ & \quad \times \Big((1 - \lambda_c)\,e_{\delta\bar{x}_3}[k] + F_{dx}[k]\,e_{\bar{a}}[k] + \varepsilon_x[k]\Big) + O(e^2) \end{aligned}$$

(e) first-order terms collected by error

$$\begin{aligned} e_{\bar{a}}[k+1] = {} & \Big(\hat{\bar{a}}'[k]\,(1 - \lambda_c) + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\Big)\,e_{\delta\bar{x}_3}[k] \\ & + \big(1 + \hat{\bar{a}}'[k]\,F_{dx}[k]\big)\,e_{\bar{a}}[k] \\ & - \frac{\hat{M}[k]}{\hat{h}[k]^2}\,e_c[k] \\ & + \frac{2\,\hat{\bar{a}}'[k]}{\hat{h}[k]}\,\hat{M}[k]\,e_{\bar{w}_s}[k] \\ & + \hat{\bar{a}}'[k]\,\varepsilon_x[k] \end{aligned}$$

Process noise and measurement noise

$$\mathbf{Q}[k] = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q_{33}[k] & -\hat{\bar{a}}'[k]\,q_{33}[k] & 0 & 0 \\ 0 & 0 & -\hat{\bar{a}}'[k]\,q_{33}[k] & \hat{\bar{a}}'[k]^2\,q_{33}[k] & 0 & 0 \\ 0 & 0 & 0 & 0 & q_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & q_{66} \end{bmatrix}, \qquad q_{55} = 0, \quad q_{66} = 0$$

$$\mathbf{R}[k] = \mathrm{diag}\big(r_1,\ r_2[k]\big)$$

2 Observability matrix

Leading order used from here on: first order in $\hat{\bar{a}}' F_{dx}$

$$\beta_k := \frac{\hat{M}[k]}{\hat{h}[k]^2}, \qquad \mathbf{F}_e[k] \to \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & 0 & 0 \\ 0 & 0 & 0 & 1 & -\beta_k & \dfrac{2\,(1-\hat{c})\,\beta_k}{\hat{h}[k]} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\text{kept:} \quad F_{e,46} = \frac{2\,\hat{\bar{a}}'\hat{M}}{\hat{h}} = \frac{2\,(1-\hat{c})\,\beta_k}{\hat{h}}$$

$$\text{dropped:} \quad F_{e,44} - 1 = \hat{\bar{a}}' F_{dx}, \qquad F_{e,43} = \hat{\bar{a}}'(1-\lambda_c) + \frac{2\,\hat{\bar{a}}'\hat{M}}{\hat{h}}$$

$$\text{steady state of the } e_{\delta\bar{x}_3} \text{ loop:} \quad (F_{e,44}-1) + F_{e,43}\Big(-\frac{F_{dx}}{1-\lambda_c}\Big) = -\frac{2\,\hat{\bar{a}}'\hat{M}F_{dx}}{\hat{h}\,(1-\lambda_c)}$$

Errors at step k_0, output errors j steps later

$$\begin{bmatrix} e_{x_1}[k_0+j] \\ e_{a_m}[k_0+j] \end{bmatrix} = \mathbf{H}\,\mathbf{\Phi}(k_0+j,k_0)\,\mathbf{e}[k_0], \qquad \beta_i := \beta_{k_0+i}, \;\; \hat{h}_i := \hat{h}[k_0+i], \;\; F_i := F_{dx}[k_0+i]$$

The matrix

$$\begin{bmatrix} e_{x_1}[k_0] \\ e_{a_m}[k_0] \\ e_{x_1}[k_0+1] \\ e_{a_m}[k_0+1] \\ e_{x_1}[k_0+2] \\ e_{a_m}[k_0+2] \\ e_{x_1}[k_0+3] \\ e_{a_m}[k_0+3] \\ e_{x_1}[k_0+4] \\ e_{a_m}[k_0+4] \\ \vdots \\ e_{x_1}[k_0+j] \\ e_{a_m}[k_0+j] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\beta_0 & \dfrac{2(1-\hat{c})\beta_0}{\hat{h}_0} \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -(\beta_0+\beta_1) & \dfrac{2(1-\hat{c})\beta_0}{\hat{h}_0} + \dfrac{2(1-\hat{c})\beta_1}{\hat{h}_1} \\ 0 & 0 & \lambda_c & -F_0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -(\beta_0+\beta_1+\beta_2) & \sum_{i=0}^{2} \dfrac{2(1-\hat{c})\beta_i}{\hat{h}_i} \\ 0 & 0 & \lambda_c^2 & -\lambda_c F_0 - F_1 & F_1\,\beta_0 & -F_1\,\dfrac{2(1-\hat{c})\beta_0}{\hat{h}_0} \\ 0 & 0 & 0 & 1 & -(\beta_0+\beta_1+\beta_2+\beta_3) & \sum_{i=0}^{3} \dfrac{2(1-\hat{c})\beta_i}{\hat{h}_i} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \lambda_c^{j-2} & -\sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_i & \sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_i \sum_{l=0}^{i-1} \beta_l & -\sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_i \sum_{l=0}^{i-1} \dfrac{2(1-\hat{c})\beta_l}{\hat{h}_l} \\ 0 & 0 & 0 & 1 & -\sum_{i=0}^{j-1} \beta_i & \sum_{i=0}^{j-1} \dfrac{2(1-\hat{c})\beta_i}{\hat{h}_i} \end{bmatrix}}_{\mathbf{O}(k_0)} \begin{bmatrix} e_{\delta\bar{x}_1}[k_0] \\ e_{\delta\bar{x}_2}[k_0] \\ e_{\delta\bar{x}_3}[k_0] \\ e_{\bar{a}}[k_0] \\ e_c[k_0] \\ e_{\bar{w}_s}[k_0] \end{bmatrix}$$

The six columns (F_{dx}, β, $\hat{h}$ constant): e_{x_1} rows, $j = 0, 1, 2, \ldots$

$e_{\delta\bar{x}_1}$	$1,\ 0,\ 0,\ 0,\ \ldots$
$e_{\delta\bar{x}_2}$	$0,\ 1,\ 0,\ 0,\ \ldots$
$e_{\delta\bar{x}_3}$	$0,\ 0,\ 1,\ \lambda_c,\ \lambda_c^2,\ \cdots \to 0$
$e_{\bar{a}}$	$0,\ 0,\ 0,\ -F_{dx},\ -F_{dx}(1+\lambda_c),\ \cdots \to -\dfrac{F_{dx}}{1-\lambda_c}$
e_c	$0,\ 0,\ 0,\ 0,\ F_{dx}\beta,\ F_{dx}\beta(2+\lambda_c),\ \cdots \approx \dfrac{F_{dx}}{1-\lambda_c}(j-3)\,\beta$
$e_{\bar{w}_s}$	$0,\ 0,\ 0,\ 0,\ -F_{dx}\dfrac{2(1-\hat{c})\beta}{\hat{h}},\ -F_{dx}\dfrac{2(1-\hat{c})\beta}{\hat{h}}(2+\lambda_c),\ \cdots \approx -\dfrac{F_{dx}}{1-\lambda_c}(j-3)\,\dfrac{2(1-\hat{c})\beta}{\hat{h}}$

The six columns (F_{dx}, β, $\hat{h}$ constant): e_{a_m} rows, $j = 0, 1, 2, \ldots$

$e_{\delta\bar{x}_1}$	$0,\ 0,\ 0,\ \ldots$
$e_{\delta\bar{x}_2}$	$0,\ 0,\ 0,\ \ldots$
$e_{\delta\bar{x}_3}$	$0,\ 0,\ 0,\ \ldots$
$e_{\bar{a}}$	$1,\ 1,\ 1,\ \ldots$
e_c	$0,\ -\beta,\ -2\beta,\ \ldots,\ -j\beta$
$e_{\bar{w}_s}$	$0,\ \dfrac{2(1-\hat{c})\beta}{\hat{h}},\ 2\,\dfrac{2(1-\hat{c})\beta}{\hat{h}},\ \ldots,\ j\,\dfrac{2(1-\hat{c})\beta}{\hat{h}}$

3 What the matrix shows

One height (β, $\hat{h}$ constant): the $e_{\bar{w}_s}$ column is a multiple of the e_c column

$$\mathbf{O}_{:,6} = -\frac{2\,(1-\hat{c})}{\hat{h}}\,\mathbf{O}_{:,5}$$

$$-\beta\,e_c + \frac{2\,(1-\hat{c})\,\beta}{\hat{h}}\,e_{\bar{w}_s} = -\beta\left(e_c - \frac{2\,(1-\hat{c})}{\hat{h}}\,e_{\bar{w}_s}\right)$$

$e_{\bar{a}}$, e_c, $e_{\bar{w}_s}$ in the e_{a_m} rows

$$e_{\bar{a}}:\;\; 1,\; 1,\; 1,\; \ldots \qquad e_c:\;\; -\sum_{i=0}^{j-1}\beta_i \qquad e_{\bar{w}_s}:\;\; \sum_{i=0}^{j-1}\frac{2\,(1-\hat{c})\,\beta_i}{\hat{h}_i}$$

Constant velocity: the two columns are what the probe has walked

$$-\sum_{i=0}^{j-1}\beta_i \approx -\int_{\hat{h}_0}^{\hat{h}_j}\frac{dh}{h^2} = \frac{1}{\hat{h}_j} - \frac{1}{\hat{h}_0}$$

$$\sum_{i=0}^{j-1}\frac{2\,(1-\hat{c})\,\beta_i}{\hat{h}_i} \approx \int_{\hat{h}_0}^{\hat{h}_j}\frac{2\,(1-\hat{c})}{h^3}\,dh = (1-\hat{c})\Big(\frac{1}{\hat{h}_0^2} - \frac{1}{\hat{h}_j^2}\Big)$$

The same two columns from the law: $\bar{a}(h_j) - \bar{a}(h_0)$ and its derivatives

$$\bar{a}(h_j) - \bar{a}(h_0) = (1-c)\Big(\frac{1}{\bar{x}_0 - \bar{w}_s} - \frac{1}{\bar{x}_j - \bar{w}_s}\Big)$$

$$\frac{\partial\left(\bar{a}(h_j) - \bar{a}(h_0)\right)}{\partial c} = \frac{1}{h_j} - \frac{1}{h_0}$$

$$\frac{\partial\left(\bar{a}(h_j) - \bar{a}(h_0)\right)}{\partial \bar{w}_s} = (1-c)\Big(\frac{1}{h_0^2} - \frac{1}{h_j^2}\Big)$$

Three heights $\hat{h}_{j_1}$, $\hat{h}_{j_2}$, $\hat{h}_{j_3}$ (e_{a_m} rows, constant velocity)

$$\det\begin{bmatrix} 1 & \dfrac{1}{\hat{h}_{j_1}} - \dfrac{1}{\hat{h}_0} & (1-\hat{c})\Big(\dfrac{1}{\hat{h}_0^2} - \dfrac{1}{\hat{h}_{j_1}^2}\Big) \\ 1 & \dfrac{1}{\hat{h}_{j_2}} - \dfrac{1}{\hat{h}_0} & (1-\hat{c})\Big(\dfrac{1}{\hat{h}_0^2} - \dfrac{1}{\hat{h}_{j_2}^2}\Big) \\ 1 & \dfrac{1}{\hat{h}_{j_3}} - \dfrac{1}{\hat{h}_0} & (1-\hat{c})\Big(\dfrac{1}{\hat{h}_0^2} - \dfrac{1}{\hat{h}_{j_3}^2}\Big) \end{bmatrix}$$

$$= -(1-\hat{c})\Big(\frac{1}{\hat{h}_{j_2}} - \frac{1}{\hat{h}_{j_1}}\Big)\Big(\frac{1}{\hat{h}_{j_3}} - \frac{1}{\hat{h}_{j_1}}\Big)\Big(\frac{1}{\hat{h}_{j_3}} - \frac{1}{\hat{h}_{j_2}}\Big)$$

c-only (0928): $\det\begin{bmatrix} 1 & \dfrac{1}{\hat{h}_{j_1}} - \dfrac{1}{\hat{h}_0} \\ 1 & \dfrac{1}{\hat{h}_{j_2}} - \dfrac{1}{\hat{h}_0} \end{bmatrix} = \dfrac{1}{\hat{h}_{j_2}} - \dfrac{1}{\hat{h}_{j_1}}$

Weight of a row pair and the part of the $e_{\bar{w}_s}$ column left after $e_{\bar{a}}$ and e_c

$$w_j = \frac{1}{r_2} + \frac{1}{r_1}\Big(\frac{F_{dx}}{1-\lambda_c}\Big)^2, \qquad \min_{\alpha_0,\,\alpha_1} \sum_j w_j \left((1-\hat{c})\Big(\frac{1}{\hat{h}_0^2} - \frac{1}{\hat{h}_j^2}\Big) - \alpha_0 - \alpha_1\Big(\frac{1}{\hat{h}_j} - \frac{1}{\hat{h}_0}\Big)\right)^2$$

Hold

$$\Delta\bar{x}_d = 0: \qquad \hat{M} = (1-\lambda_c)\,\delta\hat{\bar{x}}_3, \qquad \hat{h}_j \approx \hat{h}_0, \qquad \frac{1}{\hat{h}_{j_2}} - \frac{1}{\hat{h}_{j_1}} \approx 0$$

Height

$$\hat{h} = 22.2 \to 1.10: \quad \frac{1}{\hat{h}} = 0.045 \to 0.909, \qquad \frac{1}{\hat{h}^2} = 0.002 \to 0.83$$
$$\hat{h} = 2.0 \to 1.10: \quad \frac{1}{\hat{h}} = 0.50 \to 0.909, \qquad \frac{1}{\hat{h}^2} = 0.25 \to 0.83$$